

Acid/Base Behaviour of the Period 3 Oxides & Hydroxides (Cambridge (CIE) A Level Chemistry): Revision Note

Acid / Base Behaviour of Period 3 Oxides & Hydroxides

Period 3 oxides

- Aluminium oxide is **amphoteric** which means that it can act both as a base (and react with an acid such as HCl) and an acid (and react with a base such as NaOH)

Acidic and basic nature of Period 3 oxides table

Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂ SO ₃
Acid / base nature	Basic	Basic	Amphoteric	Acidic	Acidic	Acidic

Reaction of Period 3 oxides with acid / base table

Oxide	Chemical equation	Comment
Na ₂ O	$\text{Na}_2\text{O (s)} + 2\text{HCl (aq)} \rightarrow 2\text{NaCl (aq)} + \text{H}_2\text{O (l)}$	
MgO	$\text{MgO (s)} + 2\text{HCl (aq)} \rightarrow \text{MgCl}_2\text{ (aq)} + \text{H}_2\text{O (l)}$	Used in indigestion remedies by neutralising excess acid in the stomach

Al₂O₃	$\text{Al}_2\text{O}_3 (\text{s}) + 3\text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3 (\text{aq}) + 3\text{H}_2\text{O} (\text{l})$ $\text{Al}_2\text{O}_3 (\text{s}) + 2\text{NaOH} (\text{aq}) + 3\text{H}_2\text{O} (\text{l}) \rightarrow 2\text{NaAl}(\text{OH})_4 (\text{aq})$	Reacts with acid to form salt and water Reacts with hot, concentrated alkali to form salt
SiO₂	$\text{SiO}_2 (\text{s}) + 2\text{NaOH} (\text{aq}) \rightarrow \text{Na}_2\text{SiO}_3 (\text{aq}) + \text{H}_2\text{O} (\text{l})$	
P₄O₁₀	$\text{P}_4\text{O}_{10} (\text{s}) + 12\text{NaOH} (\text{aq}) \rightarrow 4\text{Na}_3\text{PO}_4 (\text{aq}) + 6\text{H}_2\text{O} (\text{l})$	
SO₂ SO₃	$\text{SO}_2 (\text{g}) + 2\text{NaOH} (\text{aq}) \rightarrow \text{Na}_2\text{SO}_3 (\text{aq}) + \text{H}_2\text{O} (\text{l})$ $\text{SO}_3 (\text{g}) + 2\text{NaOH} (\text{aq}) \rightarrow \text{Na}_2\text{SO}_4 (\text{aq}) + \text{H}_2\text{O} (\text{l})$	

- The acidic and basic nature of the Period 3 elements can be explained by looking at their **structure**, **bonding** and the Period 3 elements' **electronegativity**

Structure & bonding of the Period 3 oxides table

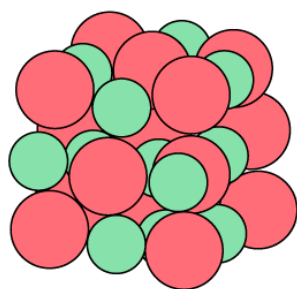
Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂ SO ₃
Relative melting point	High	High	Very high	Very high	Low	Low
Chemical bonding	Ionic	Ionic	Ionic (with some covalent character)	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent	Simple molecular	Simple molecular

Electronegativity of Period 3 elements table

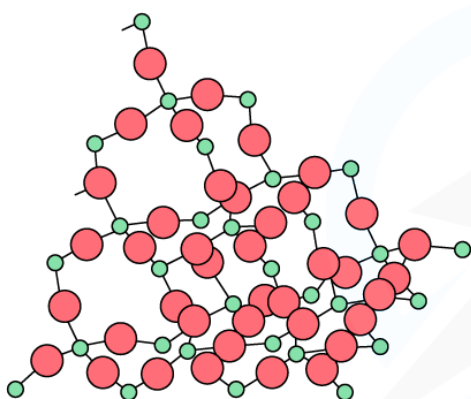
Period 3 element	Na	Mg	Al	Si	P	S	Cl
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0

- Oxygen has an electronegativity value of 3.5
- Therefore, the difference in electronegativity between oxygen and Na, Mg and Al is the largest
- So, electrons will be **transferred** to oxygen when forming oxides giving the oxide an **ionic binding**
- The oxides of Si, P and S will **share** the electrons with the oxygen to form **covalently bonded** oxides
- The **giant ionic** and **giant covalent** structured oxides will have **high melting points** as it is difficult to break the structures apart

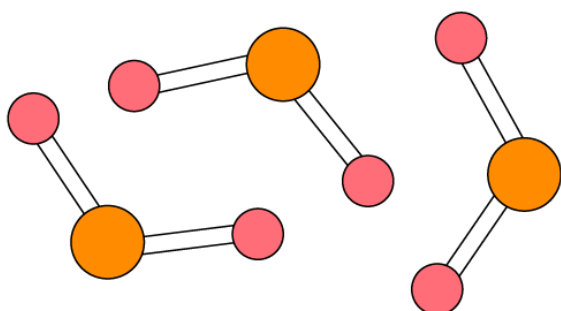
The structure of some Period 3 oxides



THE GIANT IONIC STRUCTURE
OF MAGNESIUM OXIDE (MgO)



THE GIANT COVALENT STRUCTURE
OF SILICON DIOXIDE (SiO_2)



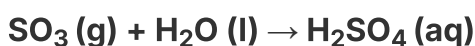
THE SIMPLE MOLECULAR STRUCTURE
OF SULFUR DIOXIDE (SO_2)

Magnesium oxide is giant ionic, silicon dioxide is giant covalent and sulfur dioxide is simple molecular

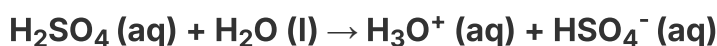
- The oxides of **Na and Mg** which show purely **ionic bonding** produce **alkaline** solutions with water as their **oxide** ions (O^{2-}) become **hydroxide** ions (OH^-):



- The oxides of **P and S** which show purely **covalent bonding** produce **acidic** solutions with water because when these oxides react with water, they form an acid which donates **H⁺** ions to water
 - Eg. SO_3 reacts with water as follows:



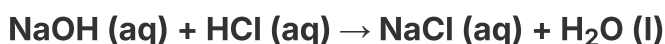
- The H_2SO_4 is an acid which will donate a H^+ to water:



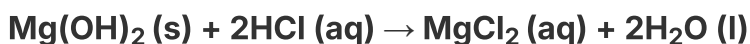
- **Al** and **Si** are insoluble and when they react with **hot, concentrated alkaline solution** they act as an acid and form a salt
 - This behaviour is very typical of a **covalently bonded oxide**
- **Al** can also react with **acidic solutions** to form a salt and water
 - This behaviour is very typical of an **ionic bonded metal oxide**
- This behaviour of **Al** proves that the chemical bonding in aluminium oxide is not purely **ionic** nor **covalent: it is amphoteric**

Period 3 hydroxides

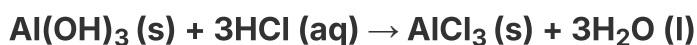
- NaOH is a strong base and will react with acids to form a salt and water:



- $Mg(OH)_2$ is also a basic compound which is often used in indigestion remedies by neutralising the excess acid in the stomach to relieve pain:



- $Al(OH)_3$ is **amphoteric** and can act both as an acid and base:





Examiner Tips and Tricks

- Electronegativity is the power of an element to draw the electrons towards itself in a covalent bond.
- For example, in Na_2O the oxygen will draw the electrons more strongly towards itself than sodium does.